MIL-C-60986A(MU)
11 May 1972
SUPERSEDING
MIL-C-60986 (MU)
11 January 1968
(See 6.12)

MILITARY SPECIFICATION

CARTRIDGE, 40MM, PRACTICE, M407A1

LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of Cartridge designated as Cartridge, 40MM, Practice, M407A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116	-	Preservation, Method of
MIL-A-2550	-	Ammunition and Special Weapons, General Specification for
MIL-P-20444	-	Primer, M42
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)

FSC: 1310

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MIL-STD-109 - Quality Assurance Terms and Definitions

MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests for

MIL-STD-1168- Lot Numbering of Ammunition

MIL-STD-1235- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

8835952 - Cartridge, 40MM, Practice, M407A1

PUBLICATIONS

ARMY

ET-8835952 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 0-190

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify, "49 CFR 0-190 (latest revision).)

3. REQUIREMENTS

3.1 Material.- Materials shall be in accordance with the applicable drawings and specifications.

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3.2 Cartridge.- The cartridge shall comply with all requirements specified on Drawing (Dwg.) 8835952, all associated drawings, and with all requirements specified in applicable specifications.

3.3 Air pressure.- The cartridge shall withstand an air pressure of three (3) pounds per square inch (psi) for five (5) seconds minimum without a drop in pressure.

3.4 Primer functioning.- The primer shall function without evidence of squibbing or hangfire. Paragraph 6.6 of MIL-P-20444B, Amendment 3 contains definitions of functioning failures.

3.5 Functioning.-

3.5.1 Cartridge.- The cartridge shall function satisfactorily and the projectile shall have a mean velocity of 247 plus or minus 7 feet per second (fps) and a standard deviation not exceeding 4.5 fps.

3.6 X-ray examination of ballistic samples.- Prior to forwarding the sample cartridges to the proving ground for ballistic testing, they shall be subjected to x-ray examination.

3.7 X-ray of fuze assemblies prior to assembling to ball and skirt assemblies. - The fuze assembly shall be x-rayed for improperly assembled, missing parts, armed or partially armed.

3.8 Production control check on loaded M118 cartridge cases. - There shall be no evidence of a primer malfunction.

3.9 Workmanship.- All parts and assemblies shall be fabricated loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required marking and stamping shall be neat and sharply defined.

3.10 Submission of results of contractor-conducted examinations and tests.- The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract (see 6.6).

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4. QUALITY ASSURANCE PROVISIONS

4.1. Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or ay other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product. - At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection.- This section is not applicable to this specification.

4.3 Inspection provisions.

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4.3.1 Lot formation. - The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision, and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall contain:

a. Cartridge case metal parts from one interfix lot number from one manufacturer.
b. Ball and skirt assemblies from one interfix lot number from one manufacturer.
c. Fuzes from one interfix lot number from one manufacturer.
d. Primers from one lot number from one manufacturer.
e. Propellant from one lot number from one manufacturer.

4.3.2 Examination.- The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects,) when so specified shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

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4.3.2.1 Cartridge case loading assembly, prior to assembling to projectile and prior to inserting base plug and primer (See Dwg. 8844609 covering a detail of Dwg. 8835952).-

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:			
1.	Powder charge cup missing	Gage	01001
2.	Spilled powder from cartridge case	Visual	01002
3.	Propellant charge weight obviously incorrect (See 6.10)	Visual	01003
	100% Inspection		
Major:			
101.	Propellant charge weight obviously incorrect (See 6.10)	Visual	01004

Minor: None defined.

4.3.2.2 Cartridge case loading assembly, prior to assembling to projectile (See dwg. 8844609 covering a detail of Dwg. 8835952).-

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Primer above flush with base of case	Gage	02008
	AQL 1.00 Percent		
Major:			
101.	Base plug above flush	Gage	02002
102.	Depth to base plug, maximum (max.)	Gage	02003
103.	Primer above flush or more than .005 below with respect to base plug	Gage Visual	02009 02005
104.	Primer damaged	Visual	02006
105.	Excessive varnish on exterior surface of primer	Automatic Gage	02010
106.	Vent hole blocked or missing		
	AQL 0.65 Percent		
Minor:			
201.	Evidence of poor workmanship (See 3.9)	Visual	02007

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4.3.2.3 Projectile, prior to assembling fuze and prior to assembling booster cup, disc and loading smoke pellet in fuze (See dwg. 8865605 covering a detail of dwg. 8835952).-

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 Percent		
	101. Booster pellet missing	Visual	03001
Minor: None defined.			

4.3.2.4 Smoke pellet (See dwg. 8862476 covering a detail of dwg. 8865605).-

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 Percent		
	101. Weight	Balance	04001
	102. Height	Gage	04002
Minor:	AQL 0.65 Percent		
	201. Diameter	Gage	04003

4.3.2.5 Projectile, prior to assembling fuze and prior to applying sealing compound on fuze threads (See dwg. 8865605 covering a detail of dwg. 8835952).-

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.65 Percent		
	101. Distance from seating surface on fuze to end of booster cup, max.	Gage	05001
	102. Gap from fuze threads to booster cup, missing	Visual	05002
	103. Smoke pellet missing	Visual	05003
Minor: None defined.			

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4.3.2.6 Projectile, prior to assembling ogive (See dwg. 8865605 covering a detail of dwg. 8835952).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Length from curved surface to top of fuze	Gage	06001
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.9)	Visual	06002

4.3.2.7 Projectile (See dwg. 8865605 covering a detail of dwg. 8835952).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Weight incorrect	Balance	07001
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.9)	Visual	07003

4.3.2.8 Cartridge, 40MM, Practice, M407A1 (See dwg. 8835952).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 1.50 Percent		
201.	Total length	Gage	08001
202.	Marking misleading or unidentifiable	Visual	08002
203.	Evidence of poor workmanship (See 3.9)	Visual	08003

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4.3.2.9 Unsealed fiber box (See dwg. 8835105).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
	AQL 0.40 Percent		
Major:			
101.	Improper number of cartridges	Visual	09001
102.	Separator missing	Visual	09002
Minor:	None defined.		

4.3.2.10 Sealed fiber box, prior to sealing bag (See Dwg. 8835105).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
	AQL 0.40 Percent		
Major:			
101.	Container damaged	Visual	10001
102.	Corners not blunted	Visual	10002
	AQL 1.00 Percent		
Minor:			
201.	Contents loose	Manual	10003
202.	Marking misleading or unidentifiable	Visual	10004

4.3.2.11 Unsealed wooden packing box (See Dwg. 8835104).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
	AQL 0.65 Percent		
Major:			
101.	Marking bag misleading or unidentifiable	Visual	11001
		Visual	11002
102.	Bag torn or perforated	Visual	11003
103.	Bag improperly sealed		
Minor:	None defined.		

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4.3.2.12 Sealed wooden packing box (See dwg. 8835104).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Box damaged	Visual	12001
Minor:	AQL 1.50 Percent		
201.	Contents loose	Manual	12002
202.	Car seal missing or improperly positioned	Visual	12003
203.	Marking misleading or unidentifiable	Visual	12004

4.3.3 Testing.-

4.3.3.1 Propellant weight of cartridge case, Code No. 13001.- The propellant weight shall be determined and then checkweighed 100 percent. The checkweighing shall be accomplished independently of the original weighing or determination, using a different balance from that used to make the original weighing, and if performed manually, shall be performed by another operator. Any charge which fails to comply with the requirement specified on dwg. 8844609, shall be classified defective and removed from the lot.

4.3.3.2 Propellant weight of cartridge case, Alternate Method, Code No. 25001.- At the start of production, three-hundred and eighty (380) consecutive samples per station, per machine shall be selected and weighed 100 percent. All samples shall comply with the applicable drawing requirement. If all samples comply with the drawing requirement, five (5) samples, per station, per every two (2) hours production shall be selected and the average weight of the five (5) samples must meet the assessed mean propellant weight within + 4 milligrams. If the average weight fails the requirement, correction will be made and three hundred and eighty (380) consecutive samples weighed from that station. If all meet the applicable requirement, the sampling every two (2) hours will be resumed.

4.3.3.3 Check test for deterioration of primer (See MIL-P-20444) Major defect (Defect Code 14001).- If the total time between original acceptance of any lot and the assembly of that lot into the cartridge exceeds two years, or if the primers have been subjected to adverse conditions, however brief, at any time since previous tests, the primer lot shall be subjected to and must satisfactorily pass,

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the check test for deterioration specified in MIL-P-20444 immediately before the primer lot is assembled into the cartridge. This test shall be performed by the contractor on primers selected by the Government Inspector at the facility assembling the primers into the cartridge (See 6.4).

4.3.3.4 X-ray examination of ballistic samples (See Paragraph 3.6), Code No. 28001.- Prior to forwarding the cartridges to the proving ground for ballistic testing, they shall be subjected to x-ray examination. If any critical defect is found, the lot shall be rejected including the ballistic samples. If any major defect is found, it shall be noted and replaced with a good sample and the samples forwarded to the proving ground. This test shall be performed as specified in 4.4.1 using equipment in accordance with 4.3.4.

4.3.3.5 Production control check on loaded M118 Cartridge Cases (See Paragraph 3.8), Major Defect, Code No. 27001.- Five (5) loaded M118 cases shall be randomly selected from every two (2) hours production from each primer assembly machine. The loaded M118 cases produced by each primer assembly machine shall be kept segregated and identified. They shall not be used until this test has been performed. If any sample fails to comply with the applicable requirement, the two (2) hours production represented by the sample shall be rejected. The test shall be performed as specified in 4.4.2.

4.3.3.6 X-ray examination of fuze assemblies prior to assembling to ball and skirt assembly (see Paragraph 3.7), Code No. 26001.- Each fuze assembly, prior to assembling to ball and skirt assembly, shall be subjected to this test. If any fuze assembly shows evidence of being improperly assembled, having missing parts, being armed or partially armed, it shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.3 using equipment in accordance with 4.3.4.

4.3.3.7 Security of crimp of base plug on cartridge case assembly (See dwg. 8844609), Major Defect, Code No. (15001) ,- Twenty (20) each of cartridge case, base plug and powder charge cup less propellant shall be selected from each lot and subjected to this test. If the crimped area of one or more assemblies fail to withstand the unseating requirement specified on the applicable drawing, the lot shall be rejected. The test shall be performed as specified in 4.4.4 using equipment in accordance with 4.3.4.

4.3.3.8 Security of crimp of ogive (See dwg. 8865605), Major defect, Code No. (16001) (when applicable).- Twenty (20) each of ogive and empty ball and skirt assemblies shall be selected from each lot. The ogive shall be cemented and crimped in place and

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subjected to this test. If the ogive of two (2) or more assemblies fail to withstand the minimum unseating requirement specified on the applicable drawing, the lot shall be rejected. The test shall be performed as specified in 4.4.5 using equipment in accordance with 4 .4.

4.3.3.9 Security of Projectile (See dwg. 8835852), Major Defect, Code No. (17001).- After assembly of the cartridge case to the projectile, eighty (80) cartridges shall be selected from each lot and subjected to this test. If four (4) or more fail to comply with the drawing requirement, the lot shall be rejected. In addition, if any one is less than 50 pounds, the lot shall be rejected. The test shall be performed as specified in 4.4.6 using equipment in accordance with 4.3.4. If the cartridge case or projectile shows any evidence of uncured sealant after completion of this test, the lot shall be rejected.

4.3.3.10 Heat seal test of packing seals (See Dwg. 8835105, Sheet 2), Major Defect, Code No. (18001 .- The samples shall be selected in accordance with Military Specification MIL-P-116. The test shall be performed as specified in 4.4.7.

4.3.3.11 Chamber gage failure (See dwg. 8835952), Major Defect, Code No. (19001),- Each cartridge after complete cure, shall be subjected to this test. If any cartridge fails to comply with the requirement specified on the applicable drawing, it shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.8 using equipment in accordance with 4.3.4.

4.3.3.12 Holding fixture vent check, Code No. (20001),- The cartridge case holding fixture used in the base plug insertion operation shall have its air vents checked per shift for any obstruction. If a vent is found to be closed or clogged, the vent shall be cleared and the previous shift's production shall be x-rayed 100 percent to insure that the propellant charge cups have not been damaged. Cartridge cases found to have damaged charge cups shall be classed defective and removed from the lot.

4.3.3.13 Air pressure (See paragraph (para.) 3.3), Major Defect, Code No. (21001).- The sampling plan for this test shall be in accordance with Military Standard MIL-STD-105, Code letter L, single plan, with an AQL of 1.5 percent. The test shall be performed as specified in 4.4.9 using equipment in accordance with 4.3.4.

4.3.3.14 Primer functioning (see paragraph 3.4), Major Defect, Code No. (22001),- Sampling shall be in accordance with Paragraph 4.2.3.2.2 of Specification MIL-P-20444B except that the sample size shall be eight hundred (800) primers from each lot. If one or more

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primers fail to comply with the applicable requirement, the lot shall be rejected. Primers not originally purchased to this requirement shall be tested for the requirement of Paragraph 3.4.1. The test shall be performed as specified in 4.4.11.

4.3.3.15 Functioning (See Paragraph 3.5), Code No. 23001.- Beginning with the first lot produced and continuing until three (3) lots have been accepted meeting the criteria of 4.3.3.15.1, the Government inspector shall select two hundred and eighty eight (288) cartridges from each lot for this test.

4.3.3.15.1 Rejection.- The lot shall be rejected if:

a. Twelve (12) or more fail to function at initial impact. (No admission of smoke shall be classified as a low failure and be placed in a dud category.) (Major defect)

b. Any primer blows back (as evidenced by perforation of the cup or extrusion of primer around firing pin causing difficulty in opening weapon.) (Major defect)

c. Any premature burst occurs (see 6.8) (Critical defect).

d. Mean velocity and or standard deviation fail to meet the applicable requirements. (Major defect)

e. Any primer misfires. (Major defect)

f. (1) Thirty (30) or more cartridge cases cannot be ejected from the launcher by spring operation of the extractor (Major defect).

(2) One (1) or more cartridge cases cannot be ejected from the launcher by hand operation of the extractor (Major defect).

g. Any projectile sticks in gun bore or fails to attain 60 percent of the mean velocity of the lot sample. (A projectile on which a velocity reading was not obtained due to low velocity shall be considered a failure). (Major defect)

h. Three (3) or more primers show any evidence of gas leakage (smoke deposit or discoloration of base plug or cartridge case base). (Major defect)

i. Any metal part separation occurs prior to impact (Major defect).

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If the launcher is suspected of being the cause of rejection for the mean velocity and standard deviation phase, sufficient cartridges from a controlled lot shall be fired from the same approved launcher. If there is significant difference in the standard deviation and mean velocity of the control lot from that previously obtained for control rounds, then a new approved launcher shall be obtained and the lot of ammunition retested. If all cartridges function properly on original test, then only the velocity will be considered on retest. If there is no significant difference in the standard deviation or mean velocity of the control lot, then the lot shall be rejected.

4.3.3.15.2 Suitability test plan (informational test only), Code No. 23002 .- Beginning with the first lot produced by each producer and continuing until that producer has submitted three (3) consecutive lots that are accepted in accordance with Paragraph 4.3.3.15, an extra sample of ninety (90) rounds shall be selected from each lot and preconditioned and tested as follows:

a. The ninety (90) rounds shall be packaged in their regular shipping containers and subjected to transportation vibration environment in accordance with MIL-STD-331, Test No. 104.

b. After transportation vibration, the rounds shall be unpacked and subjected to the JAN temperature and humidity cycle (14 days) specified in MIL-STD-331, Test No. 105.

c. After the temperature and humidity cycle, the rounds shall be temperature conditioned and fired as follows:

30 rounds +165 F ± 5 degrees F.
30 rounds ambient (between +40 and +110 degrees F).
30 rounds -65 degrees F ± 5 degrees F.

d. Fire the rounds in accordance with Paragraph 4.4.10 and with TECOM Material Test Procedures, TECP-700-700 Volume 4. Test results, failure analysis reports, and other information, as requested shall be forwarded to Picatinny Arsenal, Attn: SMUPA-QA-A-S for evaluation.

4.3.3.16 Functioning (See Paragraph 3.5), Major defect, Regular Production, Code No. (24001).- After three (3 acceptable lots select seventy-two (72) cartridges from each lot for this test. If two (2) or more lots fail the criteria of 4.3.3.16.1, the consecutive lots thereafter shall revert to the criteria of 4.3.3.15.

4.3.3.16.1 Rejection.- The lot shall be rejected if:

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a. Four (4) or more fail to function at initial impact. (No admission of smoke shall be classified as a low order failure and be placed in a dud category.) (Major defect)

b. Any primer blows back (as evidenced by perforation of the cup or extrusion of the primer around firing pin causing difficulty in opening weapon. (Major defect)

c. Any premature burst occurs (see 6.8) (Critical defect).

d. Mean velocity and or standard deviation fail to meet the applicable requirement (Major defect).

e. Any primer misfires (Major defect).

f. (1) Five (5) or more cartridge cases cannot be ejected from the launcher by spring operation of the extractor. (Major defect)

(2) One (1) or more cartridge cases cannot be extracted from the launcher by hand operation of the extractor (Major defect).

g. Any projectile sticks in the gun bore or fails to attain 60 percent of the mean uncorrected velocity of the lot sample. (A projectile on which a velocity reading was not obtained due to low velocity shall be considered as a failure. (Major defect)

h. Three (3) or more primers show any evidence of gas leakage (smoke deposit or discoloration of base plug or cartridge case base). (Major defect)

i. Any metal part separation occurs prior to impact (Major defect).

If the launcher is suspected of being the cause of rejection for the mean velocity and standard deviation phase, sufficient cartridges from a controlled lot shall be fired from the same approved launcher. If there is significant difference in the standard deviation and mean velocity of the control lot from that previously obtained for control rounds, then a new approved launcher shall be obtained and the lot of ammunition retested. If all cartridges function properly on original test, then only the velocity will be considered on retest. If there is no significant difference in the standard deviation or mean velocity of the control lot, then the lot shall be rejected.

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4.3.4 Inspection equipment.- Equipment Tabulation (ET number 8835952 and Equipment Lists (EL's) referenced thereon identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3. Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

4.3.4.1 Government rights to documentation.- Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, contruction, survey, and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures.

4.4.1 X-ray examination of ballistic samples.-The cartridge, prior to forwarding to the proving ground, shall be subjected to x-ray examination using the x-ray equipment specified on the Equipment List or any other method satisfactory to the Contracting Officer. This test is a nondestructive test. Parts so tested may be returned to the lot.

4.4.2 Production control check on loaded M118 cartridge cases.- The loaded M118 cartridge case shall be fired from an approved launcher placed on an approved mount. Observation shall be made for primer functioning. This test is a destructive test. Parts so tested shall not be returned to the lot. NOTE: No projectile is required for this test.

4.4.3 X-ray examination of fuze assemblies prior to assembling to ball and skirt assembly.- Improperly assembled, missing parts, armed or partially armed fuze assemblies shall be determined by x-ray equipment or any other method satisfactory to the Contracting Officer. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.4 Security of crimp of base plug on cartridge case assembly.- The assembly shall be placed in the fixture shown on the Equipment List and the minimum unseating force specified on Dwg. 8844609 shall be applied to failure. This test is a destructive test. Parts so tested shall not be returned to the lot.

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4.4.5 Security of crimp of ogive.- The assembly shall be placed in the fixture shown on the Equipment List and the minimum unseating force specified on Dwg. 8865605 shall be applied. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.6 Security of projectile.- The cartridge shall be placed in the fixture shown on the Equipment List and the axial force specified on Dwg. 8835952 shall be applied. Projectile must separate from the cartridge case in the range specified on Dwg. 8835952. This test is a non-destructive test. Parts so tested may be returned to the lot. Stake shall be removed from the loaded case prior to returning to the lot.

4.4.7 Heat seal test of packing seals.- This test shall be performed in accordance with MIL-P-116. Samples so tested shall not be returned to the lot.

4.4.8 Chamber gage failure.- The cartridge shall be placed in the chamber shown on the Equipment List and must meet the requirement specified on Dwg. 8835952. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.9 Air pressure.- The cartridge shall beplaced in the fixture shown on the Equipment List and a measured quantity of air shall be applied to produce the required air pressure. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.10 Functioning.- This test shall be conducted at a Government owned proving ground. The cartridge shall be fired from an approved launcher placed on an approved mount to impact at 200 meters minimum from the launcher on a six inch thick loose gravel pad. The gravel used shall have a 1/2 inch maximum size. The velocity shall be measured at 20 feet from the muzzle of the launcher, using two sets of screens, each set of screens having an electronic counter. One set of screens shall be placed 14 and 26 feet from the launcher and the other set 15 and 25 feet from the launcher. NOTE: To insure that the chronographs display accurate time intervals, the ratio of the two times shall not differ from 1.200 by more than plus or minus 0.005. If the ratio differs from 1.200 by more than plus or minus 0.005, the velocity results for that round shall be discarded and excluded from the calculation of mean velocity and velocity standard deviation. (No more than four (4) rounds in a sample size of 72 or twelve (12) rounds in a sample size of 288 may be disregarded.) If the applicable number of improper ratios is exceeded, additional round(s) must be fired Observation shall be made and

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results recorded for functioning, mean velocity and standard deviation (See 6.7). The mean velocity of the lot being tested shall be corrected by the correction factor determined for the particular test launcher being used on the day of the test firing by firing control rounds in accordance with Table I. When velocity values are suspected to be outliers, they shall be tested, using the statistical procedure approved by the Government proving ground and the mean velocity and standard deviation adjusted accordingly. However, no more than four values consisting of no more than two on the high side and no more than two on the low side, shall be considered for exclusion.

TABLE I (See 6.11)

Standard Deviation For Control Lots	Number of Rounds
3 or less	24
3.1 - 3.5	36
3.6 - 4.0	60
4.1 - 4.5	84

4.4.10.1 When any one sample lot from a regular ammunition lot is broken into two (2) groups and fired separately, the following procedure shall be followed when calculating mean velocity and standard deviation.

1. Individual velocities shall be corrected for each group using the correct factor of that group.

2. The mean velocity and standard deviation shall be calculated from all the correced velocities.

4.4.10.2 Test Validity.- If, for any reason, proving ground considers that the test conditions have detrimentally affected the test results, the test shall be declared invalid and a new test shall be performed with additional samples.

4.4.11 Primer functioning.- Each primer of the 800 samples shall be tested as specified in MIL-P-20444B, Paragraph 4.3.2.2.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging.-

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5.1.1 Level C.- Preservation and packaging shall be in accordance with dwg. 8835105.

5.2 Packing.-

5.2.1 Level C.- The cartridge shall be packed in accordance with Dwg. 8835104.

5.3 Marking.- Marking shall be in accordance with Dwgs. 8835105 and 8835104 and Code of Federal Regulations, Title 49, Parts 0-190.

5.4 Shipping.- When components from more than one lot are shipped as a carload each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data.- Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers.- The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. These code numbers shall be listed on drawings of equipment designs submitted by the contractor (see 6.3.5).

6.3 Inspection equipment - The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Equipment lists.- Equipment lists indicate the availability of inspection equipment designs by showing in the "Part Number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as (Contractor Design" in the "Part Number" column. These contractor

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designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Equipment list codes.- The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.- Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: Mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references mandatory Army Designs, the contractor shall comply with, and use, these designs accordingly. The contractor may, however, in connection with non-mandatory designs, with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor-prepared designs (see 6.3.4) or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.- Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category A, Form 3. Legibility and reproducibility shall permit conventional making of clearly understandable, high-contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

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6.3.5 Submission of contractor design.- All submitted designs shall contain a reference to the applicable 5-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, Attn: SMUPA-QA-A-S. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally, within one month after receipt.

6.4 Cost of check test.- The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test for deterioration of the primer assemblies. The tests shall be conducted at Government expense without cost to the contractor who loaded the primer assemblies or to the contractor assembling the primer assemblies into the cartridge, and shall not constitute a basis for rejection against either contractor except where deterioration has occured as a direct result of carelessness in handling, storage, etc., permitted while the primer assembly lot was under the jurisdiction of either contractor.

6.5 Submission of test data.- In addition to the normal distribution of records, when the cartridge is procured by the U.S. Army Munitions Command, one (1) copy of all ballistic data and ammunition data cards shall be forwarded to: Commanding Officer, Picatinny Arsenal, Attn: SMUPA-QA-A-S.

6.6 Submission of results of contractor-conducted examinations and tests.- Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, Attn: SMUPA-QA-A-S.

6.7 Standard deviation shall be calculated from the following formula: (Other approved standard deviation formulas may be used.)

$$s = \sqrt{\frac{\leq (x-\bar{x})^2}{n-1}}$$

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where: X = each individual value

$\bar{X}$ = sample arithmetic mean $\frac{\Sigma X}{n}$

n = sample size

$(X - \bar{X})$ = the sum of the squares of the differences between the sample mean and each individual value.

6.7.1 For a faster and easier method of arriving at the standard deviation formula, the following alternate method may be used:

$$S = \sqrt{\frac{n\Sigma X^2 - (\Sigma X)^2}{n\ (n-1)}}$$

6.8 Premature burst.- A premature burst is considered to have occurred if the round functions in the gun bore or in flight.

6.9 Combining of proving ground tests.-When the contractor for the cartridge is also the contractor for one or more of the components thereof, the proving ground tests of the contractor may be combined with the proving ground tests of the cartridge, to save expense, upon agreement between the procuring activity and the contractor. In cases where the cartridge specification does not cover all of the proving ground tests specified for the component, the additional tests specified in the component specification shall be conducted.

6.10 If any cartridge case, prior to insertion of the base plug and primer is suspected of containing an underweight or overweight propellant charge, based on visual inspection it will be removed from the lot and weighed on suitable scales. Any cartridge case found with less than 75 percent or more than 125 percent of the assessed propellant load will be classified as a critical defect and removed from the lot. Any cartridge case found to have a propellant weight out of drawing tolerance but within ± 25 percent of the assessed propellant load will be classified as a major defect and removed from the lot.

6.11 Table 1 shall be used at the loading plants for powder assessment.

6.12 Supersession data.- This specification includes the requirements of, and supersedes, Purchase Specification AEI-63F, dated 1 August 1966.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1310-A-019

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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